

JAYANT KUSHWAHA

M.Sc. BIOTECHNOLOGY

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MSc Biotechnology student, dissertation at ICMR-NICPR on cervical cancer genomics. My published work (Virus Genes, 2025) looked at inhibiting Monkeypox virus proteins through computational screening PPI networks, molecular docking, enrichment analysis. On the wet-lab side, I've done DNA extraction, ELISA, and SDS-PAGE. I want to keep working at the crossover between genomics and computation, ideally in cancer biology or infectious disease research.

EDUCATION

MASTERS OF SCIENCE IN BIOTECHNOLOGY

2024 - 2026

• NOIDA INTERNATIONAL UNIVERSITY

C.G.P.A – 9.10

- **Dissertation:** Systems-level Analysis Of Cervical Cancer Related Dysregulated Genes Using Computational Approaches

BACHELORS OF SCIENCE IN BIOTECHNOLOGY

2021 - 2024

• R.N.C Arts J.D.B Commerce N.S.C Science College Nashik.

C.G.P.A – 8.66

- **Dissertation:** Synthesizing Zinc oxide Nanoparticles from Banana peels and its applications.

EXPERIENCE

MASTER'S DISSERTATION

Jan – 2026 to April - 2026

ICMR - NATIONAL INSTITUTE OF CANCER PREVENTION AND RESEARCH.

- Curated literature-derived gene expression data from cervical cancer studies and compiled a dysregulated gene set for downstream computational analysis.
- Ran functional enrichment via Enrichr, mapping the gene set to oncogenic pathways.
- Built protein-protein interaction networks in STRING and visualized them in Cytoscape to identify central network nodes based on degree and betweenness centrality.
- AKT1 and EGFR emerged as hub genes both are well-established in cervical cancer biology, which adds confidence to the network model.

INTERNSHIP

23rd Jan – 28th Jan 2023

AGRI SEARCH (INDIA) PVT LTD.

- Acquired knowledge about manufacturing process carried out in Agri Search and obtained understanding of water assessment.
- Studied Quality Control and Quality Assurance techniques required in industries.
- Experienced how the fermentation processes is done.
- Observed how the Bio-fertilizers are prepared and tested.

SKILLS

Bioinformatics & Computational:

- **Structure & Docking:** AlphaFold2, AutoDock Vina, PyMOL, Discovery Studio, PDB, Maestro - Schrödinger.
- **Network & Pathway Analysis:** STRING, Cytoscape, Enrichr
- **Sequence & Database Tools:** NCBI, BLAST, GenBank, PubChem

Wet-Lab Techniques:

- **Nucleic Acid Techniques:** DNA extraction, PCR, Gel Electrophoresis, Restriction Digestion
- **Protein Techniques:** SDS-PAGE, Protein Estimation, ELISA
- **Cell & Microbial Techniques:** Bacterial Transformation, Competent Cell Preparation, Plant-Tissue Culturing

Programming: Python (BioPython), C/C++

PROJECTS

- **Structure-Based Virtual Screening of Inhibitors Against Monkeypox Virus Proteins**
Independent Research Project – 2025
 - Modeled MPXV proteins using AlphaFold2.
 - Retrieved and prepared ligands from PubChem for docking.
 - Performed virtual screening using AutoDock Vina.
 - Analyzed protein-ligand interactions with PyMOL and Discovery Studio.
- **Synthesis of Zinc oxide Nanoparticles and its Application**
Undergraduate Dissertation Project -2024
 - Synthesized zinc oxide nanoparticles from Banana peels extract by bottom-up approach.
 - Characterization by UV-Vis Spectrophotometer, FTIR, SEM.
 - Tested the Antimicrobial-activity assay on *E.coli* and *S.aureus*.
 - Applied on Mung plants and studied their chlorophyll and Protein content.

PEER-REVIEWED PUBLICATIONS

- Singh, S., Sharma, V. K., Shrivastav, D., **Kushwaha, J. M.**, Mishra, M. K., & Ali Beg, M. M. (2026). From Rhizosphere to Resistance: Microbe-Plant Interactions in Eco-Smart Biocontrol. *MicrobiologyOpen*, 15(5), e70398.
<https://doi.org/10.1002/mbo3.70398> **I.F. 3.6**
- Kumari, R., **Kushwaha, J. M.**, Sharma, V. K., Shrivastav, D., & Singh, S. (2026). Distinct molecular pathways driving lung cancer in non-smokers: Etiology, signaling networks, and clinical implications. *Critical Reviews in Oncology/Hematology*, 226, 105436.
<https://doi.org/10.1016/j.critrevonc.2026.105436> **I.F. 6.2**
- **Kushwaha, J.M.**, Rani, M.S.S. & Singh, S. Targeting monkeypox virus (MPXV): strategies for molecular docking studies on protein inhibition. *Virus Genes* (2025).
<https://doi.org/10.1007/s11262-025-02171-1> **I.F. 1.9**

BOOK CHAPTERS

- Singh S.*, **Kushwaha, J.M.** (2026). "Next-Generation Biotechnology for Jaivik Kheti: Biofertilizers & Plant Growth Promoters." In Organic Agriculture and Sustainable Farming Systems (Eds. Hasan F., Singh A.), pp. 15-19. Learning Media Publication, Meerut. ISBN 978-81-69199-14-8.

CERTIFICATES

- **Bioinformatics Internship** - SNP data analysis and AI genomics prediction model building (5-day, Feb 2025)
- **Focus on Peer Review** - Nature Masterclasses; manuscript evaluation and reviewer standards (*Jul 2025*)
- **ELISA Workshop** - Quantitative TSH estimation in human plasma, hands-on (*Mar 2025*)
- **C/C++ Programming** - Disha Computers (*Aug 2023*)
- **LifeArc Biology Research Job Simulation** - Forage, differentiation protocol optimization (*Jan 2025*)
- **Tata Data Visualization Job Simulation** - Forage, executive-facing data visualization (*Aug 2024*)

PROFILES

-  **LinkedIn-** [Jayant Kushwaha](#)
-  **ORCID ID-** [0009-0000-8320-9034](#)
-  **WOS ID-** [NVM-0932-2025](#)

-  **Scopus ID-** [59959633300](#)
-  **Research Gate ID-** [Jayant M. Kushwaha](#)
-  **Github-** [jayant10014](#)